"Weight Loss Journey Is Not Alone": Designing an Anonymous Chat System with Two Collaborative AI Agents to Foster Social Support for Women Using GLP-1s

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Women using GLP-1 for weight loss seek to share experiences and find companionship in safe group chats. As generative AI enters online multiparty interactions, supporting group conversations in online health communities becomes a key design challenge. Based on a three-stage formative study, we designed an anonymous group-chat system. This study is the first to examine a previously unexplored "dual-AI complementarity" mechanism in existing online health social settings. We conducted a 16-day controlled study with 24 women. The AI-supported group showed more active information seeking, relationship building, experience disclosure, and group-chat participation, while the private AI provided personalized companionship and a low-pressure space for expression. The distinctive value of the dual-AI design lies in enabling support to flow between private expression and group communication while giving users control over this flow. This study informs the coordinated design of private and group-chat AI in online health social contexts.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**; *Human computer interaction (HCI).*

Additional Key Words and Phrases: multi-AI interaction, online health communities, anonymous peer support, group-chat AI, private AI, social support, GLP-1

1 Introduction

Obesity remains a major global public-health challenge. As a chronic, relapsing disease shaped by genetic, behavioral, and environmental factors, it requires long-term management [6, 24, 98]. HCI research has examined how digital technologies become embedded in everyday weight management. One line of work investigates self-management mechanisms—including data monitoring, goal setting, and feedback—and how they support users in reflecting on and adjusting weight-loss behaviors [3, 6, 37, 97, 142]. Another focuses on social-support mechanisms—including online communities, peer support, and partner systems—and how they facilitate experience sharing and reciprocal encouragement [34, 49, 51, 63, 71, 93, 100, 146].

Over the past three years, GLP-1, particularly semaglutide and tirzepatide, have attracted widespread attention because of their substantial weight-loss effects and have generated a large global user population [14, 52, 95, 102]. These medications primarily promote weight loss by suppressing appetite and reducing energy intake. Users therefore no longer rely solely on willpower to control their diets, challenging traditional approaches centered on diet and exercise [14, 95]. Increasing numbers of GLP-1 users now discuss medication effects, side effects, lifestyle adjustments, and medication-related stigma through online channels. In this context, social support provides an important lens for understanding and supporting GLP-1-assisted weight loss [61, 102, 147].

HCI researchers have increasingly examined how online health-support technologies and social-support mechanisms shape communication and create design opportunities [18, 19, 47, 61, 84, 101, 108, 138]. Prior work has addressed expression and privacy [5, 32, 53, 55, 96, 141], judgment of the credibility of information [53, 96], peer matching and group stability [52, 75, 129, 138, 147], and the timeliness and continuity of responses [75, 129, 138]. These concerns are especially salient for GLP-1-supported weight loss. Variation and uncertainty in treatment effects and side effects can make online exchanges vulnerable to the misapplication of personal experience, negative judgment, and medication-related stigma [14, 95, 96]. Research focused on women further suggests that GLP-1 use attracts stronger negative

judgment than diet- and exercise-based weight loss because it is framed as "taking a shortcut" [5, 32, 55]. Such judgment can discourage women from seeking or providing support online. Together, these concerns motivate the design of a safe and trustworthy peer-support space for women using GLP-1 [32, 101, 120].

Accordingly, we ask:

RQ1: What new digital support needs do women using GLP-1 for weight management have?

RQ2: What interaction mechanisms and design approaches can better support safe and trustworthy sharing within this group?

Conversational AI based on large language models has been applied to digital weight management. Existing research typically positions AI as a personal health coach, behavioral prompt, or facilitator of reflection [23, 75, 114]. It examines how role positioning, personalization, interactional initiative, anthropomorphism, and persuasive language support individual weight regulation. However, few studies have examined how AI can be embedded in communication within weight-loss groups or how it affects reciprocity and social support. Other studies have explored AI roles in group interaction, including replacing human responses [45, 76, 79], helping users draft replies [82, 129], and coordinating turn-taking and participation balance [28, 29, 43, 60, 117]. This work often focuses on one-off conversations or non-health contexts. How AI should enter an ongoing online health-support process therefore remains unclear.

We therefore further ask:

RQ3: How should AI intervene and be positioned in group interactions to support social support among women using GLP-1?

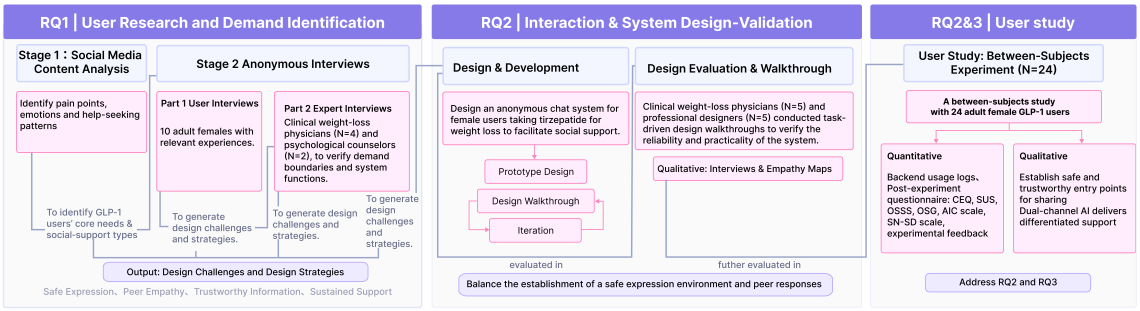


Fig. 1. Overall structure and logic of the papers

Figure 1 shows the structure of the study. To address the research questions, we first conducted a formative study comprising (1) social-media data analysis and (2) semi-structured interviews with 10 GLP-1 users and 6 experts. We examined the digital peer-support needs and expressive concerns of women using GLP-1 for weight loss (Figure 1, RQ1). Participants preferred discussing bodily changes, side effects, and weight-loss anxiety with unfamiliar peers who had similar experiences. They also wanted a safe, trustworthy, and sustained environment and the ability to choose forms of support that suited their needs.

Guided by these findings, we designed and implemented Mira, an AI-supported anonymous peer-support system for women using GLP-1 for weight loss. We then conducted a two-week user study with 24 adult women. Questionnaires, backend interaction logs, and post-study interviews were used to examine experience sharing and perceived social support and to analyze how AI entered group chat and was understood within group interaction. Relative to the non-AI condition, the AI-supported condition showed a mixed pattern: AI helped sustain some exchanges and was associated

with more positive companionship and social-support experiences, while also revealing limits in timing, interpretation, and intervention. We conclude by discussing these findings, limitations, and directions for future research.

This study makes four contributions: (1) it identifies the emotional, cognitive, and social-support needs of women using GLP-1 for weight management; (2) it proposes interaction strategies that use anonymous group chat to support the sharing of sensitive bodily experiences and peer support; (3) it designs and implements a system that explores AI's role in carrying forward topics and emotions and organizing collective experience; and (4) it uses quantitative and qualitative evidence to examine the feasibility of AI as a boundary mediator in weight-loss peer support, offering implications for anonymous digital support systems.

2 Concepts Background and Positionality

2.1 GLP-1 for Weight Management

GLP-1 receptor agonists were initially used to manage blood glucose in people with type 2 diabetes and later entered weight-management practice. These medications can produce substantial and sustained weight loss in adults with obesity or overweight [135]. Unlike approaches that rely primarily on dietary control and exercise plans, GLP-1 do not depend entirely on conscious willpower. They manage weight by slowing gastric emptying, increasing satiety, reducing appetite, and regulating energy balance [33, 135]. For users, weight loss is a long-term process that can involve adapting to side effects, shame, anxiety, and changes in self-identity. FDA labeling indicates that common adverse effects of tirzepatide include nausea, diarrhea, vomiting, constipation, dyspepsia, and fatigue [33]. Users actively share highly individual side effects and medication-related uncertainties in online communities [13].

2.2 Social Context of Medication-Assisted Weight Loss

The Global Burden of Diseases Study (GDB) reports that nearly one third of the global population is overweight or obese [90]. In recent years, there has been rapid spread of GLP-1 drugs and market expansion. FF National survey shows that 18 per cent of adults in the United States have used GLP-1 drugs and 12 per cent are currently using drugs; women currently use more than men, and 22 per cent of non-participants are interested in reducing their weight [85]. These data show that the GLP-1 has expanded from a clinical programme for a small number of patients to an important component of mass weight management. In China, the Guidelines for Obesity Care in China, published in 2024, have included GLP-1RAs such as semaglutide in the treatment programme for obesity drugs, indicating that such drugs have entered Chinese practice to regulate the management of chemical weight and that prescription and drug procurement data indicate a rapid increase in clinical use of GLP-1RA in China [36]. At the same time, the prevalence of the GLP-1 has led to a discussion of physical, identity, medical accessibility and social inequality. The GLP-1 receptor agonists are not only biomedical innovations but are also influencing physical norms, health systems and societal attitudes.

Another relevant foundation is Feminist HCI, which centers the agency, lived conditions, and social identities of women and marginalized groups. It calls for design to respond to embodied experience while critically examining technology's effects. For women using GLP-1, side effects, bodily changes, medication-related stigma, and judgments in interpersonal, community, and broader social contexts make weight-loss experiences diverse and situated.

2.3 Researcher Positionality

Women on our research team had previously used this medication on a physician's advice. This experience sensitized us to subtle aspects of side effects, bodily changes, and social judgment. We treated it as an entry point into the research context, while grounding the system design in formative research, expert feedback, and user evaluation.

3 Related Work

3.1 Digital Weight-Management Tools and Long-Term Health Management

The weight loss is often experienced as a lonely and self-restraint process [6, 24, 98]. Until GLP-1 drugs are widely used, diets and sports have long been the dominant mode of weight loss [30, 98, 122]. Existing HCI research in the area of weight loss has broadly developed two interrelated pathways: one focusing on how individuals can record and understand body changes using digital tools, focusing mainly on data recording, feedback presentation, target setting, personalization, self-control, etc. [3, 6, 37, 41, 42, 67, 97, 103, 110, 118, 124, 142]. This resulted in the emergence of various types of tools, including health apps, smart scales, fitness trackers, and wearable devices [41, 42, 67, 97, 103, 118]. Another focuses on how participants understand physical changes together with their peers and adjust daily weight loss practices. For example, the comparison of objectives, the demonstration of progress, social incentives, self-disclosure and the impact of membership on social support [34, 51, 63, 71, 93, 100, 113], the visibility of records, how to respond and how to collate experience, understanding of data, comparison and continuation of discussions [80, 86, 124, 142]. social support in the online digital channel is usually encouraged, advised, shared experience, commented, praised and quoted chat content [47, 101, 108]. The members therefore engaged in an empirical discussion around the outcome of the phase [91, 92], the experience of failures and setbacks [92, 101], and the objectives and behaviour of weight loss [49, 146]. In general, however, both of these research links are essentially exploring how technology can help participants to listen to their own bodies.

In long-term health management, obesity has chronic and re-emergible characteristics and long-term maintenance requires changes in daily living habits [39], including start-up, adjustment programmes, maintenance and re-participation after interruption [73, 77, 116]. Thus, digital products need to reduce the recording burden, provide trend feedback to help participants understand changes, preserve historical progress and provide an entry point for re-engagement in weight loss [3, 6, 15, 106]; interactive design should also support different phases of operational adjustments and peer responses [70, 121, 124].

The use of GLP-1 continues to characterize long-term weight loss management, but the content shared by participants has changed. In addition to weight and diet, participants discuss injection stages, appetite, physical reactions, side effects, dosages and stigma [14, 95, 134], but these experiences are phased and individual differences [14, 95].

3.2 Social Support in Online Health Communities

Online health social networking places health management in continuous relationship interactions, with participants sharing experiences, responding to emotions and obtaining advice through postings, comments and group exchanges. In this context of interaction, the process by which individuals receive emotional, information, evaluation, belonging and practical assistance from social relations is known as social support [18, 46, 64, 111], usually divided into four dimensions of emotional, information, evaluation and instrumental support [46]. The study initially focused on cancer survivors and rehabilitation training [7, 40], and then expanded to mental health [17, 107], XXX relax training [2, 107] and routine health scenarios such as scabies and weight loss [68, 88].

Digitalization has expanded the interactive approach of social support [38, 84]654 and interaction with the personal perspective of a wide range of online healthy societies contributes to social bonding among peers and complements the less involved physical feelings in formal health services 654 [36]. Further attention is being given to how different modes and mechanisms of interaction can contribute to the generation of such support in the HCI and CSCW areas. The online channel provides informational support [19, 84] mainly through advice and experience exchange, emotional support [19, 99], through mutual interest and encouragement, and supports respect and networking with recognition and common identity [18, 111]. Among them, the Forum relies on the sustainability of content to accumulate searchable empirical knowledge (654). The group chat improves the timeliness of support through continuous responses [129, 138]. Sharing progress allows peers to understand each other's daily changes and to provide timely feedback [78, 127].

social support in the reduced scene is mainly structured around body weight progress, routine response and behaviour maintenance [21, 51, 119]. Experience-sharing in the weight-loss forum helps peers understand situations, identify feelings and provide practical experience [101]. Research has been carried out to promote interaction among weight-earners using online channel [146]). For example, the WSHS system encourages participants to regularly share and view progress in weight loss, which has a better short-term weight loss than private nutritionist counselling [49]. In addition, social support has evolved with the weight loss phase. During the start-up period, members can gradually become familiar with groups and remain engaged by documenting goals and progress, participating in common challenges, viewing group achievements and interacting with peers [10, 127, 146]. In the continuing phase, non-judgmental responses and similar experiences can help members gain practical experience and sustain [21, 101], commending peers and helping others to sustainably bring social support and momentum [58, 127].

Online GLP-1 weight-management programs place sharing and exchange within ongoing medication use and behavior management [102]. Comparing treatment progress can help users assess whether weight changes and side effects are common, while stable membership can support trust and experience sharing [52]. However, an individual medication narrative may be mistaken for a generalizable experience and influence others' medication decisions [96].

3.3 Generative Conversational AI in Group Interaction

In recent years, large language models have been widely used in conversational systems [65, 128] and have entered settings such as health-behavior interventions and psychological support [65, 75, 128, 145]. In health contexts, researchers have begun to position AI as an aid to interpersonal support that helps peers and caregivers express support more effectively. One stream addresses individual users, including daily planning for people with depression [87], smoking-cessation reminders [57], and answers to health questions [140]. Another examines how AI intervenes in multiparty interaction [28, 45, 76, 79], including topic advancement [43, 117], participation [28, 29, 43, 60, 117, 126], intervention timing and response targets [45, 76, 79, 133], dialogue-state analysis [79], role positioning [56, 62, 74], linguistic style [125, 144], and group-experience synthesis [69, 136].

With these issues in mind, the study identified as a different actor in multi-person interactions: first, the moderator. The discussion was advanced through the allocation of statements, control of the rhythm and concluding observations [43, 60, 117, 126], such as the Groupfeed Bot system, which promoted balanced participation by medium-sized group members by controlling the discussion time, inviting inactive members to speak and collating their views. Second, information broker. Information intermediaries generate summaries of group content, extract views of different members and generate follow-up leads [69, 133, 136]. Third, Proactive participants. AI agents select interventions or invite members based on the state of dialogue and allow members to adjust their frequency of participation [45, 76, 79]; and fourth, facilitators. In case of disagreement or miscalculation by AI, AI leads to the rotation of expressions, offers

compromise and wrong amendments [29, 104, 133]. For example, AI Chatbot helps to create a safer communication environment [104] by mediating multi-person privacy conflicts, leading consultations, reducing offensive expression and offering compromise solutions.

Current AI applications in the area of weight loss continue to focus on one-on-one interactions with individual participants: systems record weight, diet and activity data and provide personalized advice, motivational information and emotional support [23, 75, 114]. WaznApp offered immediate support based on the progress of participants, but is still in the programme phase [12]. Bloom uses LLM trainers, progress summaries and proactive announcements for daily campaign support. Studies for overweight participants show that clear and supportive expression increases the willingness to use pleasure and follow recommendations [22].

4 Formative Study

This study was approved by the local science and technology ethics review committee. All participants provided informed consent. We used anonymous online communication throughout the study and did not disclose questionnaire or interview content to group members, reducing concerns about identity exposure, group pressure, and evaluation by others.

4.1 Stage 1: Social-Media Retrieval and Content Analysis

4.1.1 Methods and Procedure. To understand GLP-1 users' discussions in naturalistic settings, we retrieved public posts and comments from Rednote using the keywords specified in the Chinese source. We standardized the title, body, tags, time, and comment fields and assessed relevance from the context of each post and its comments. We excluded duplicates, incomplete entries, advertisements and purchasing agents, medical-aesthetic promotions, non-GLP-1 weight-loss content, and content without authentic medication-use experience [1, 25, 27, 132]. After cleaning, we retained 463 posts and 29,599 comments. Contextual review yielded 444 eligible posts and 28,529 comments. We classified themes at the post and comment levels and generated analytic labels using keywords, rules, and contextual interpretation.

4.1.2 Findings. As a result of the thematic clustering and analysis of the data, the members of the study summarized the discussion as one of several topics: first, the path to drug use and the physical changes it brings; second, why participants share experiences and how to communicate online; and third, the need to reduce stigma and identity protection associated with medication use. The semi-structured interviews on the subject of phase II provided leads.

4.2 Stage 2: Anonymous User Interviews

4.2.1 Participants. Using semi-structured interviews, the study takes into account the experience of the GLP-1 adult female participants and clinical experts to gain insight into the design challenges of digital weight loss systems in terms of the real needs of participants, the mechanisms of interaction and the medical safety boundaries. A total of 16 participants were recruited. These include 10 female people using GLP-1, 4 obese medical specialists and 2 clinical psychologists.

Table 1. Population information tables for interviews

Serial number	participants number	Gender	Time used
1	P1	Women	1 month
2	P2	Women	5 weeks
3	P3	Women	About 8 months
4	P4	Women	About a month and a half.
5	P5	Women	1 year
6	P6	Women	6 months
7	P7	Women	About 3 months
8	P8	Women	1 month
9	P9	Women	About 4 months
10	P10	Women	2 months

Semi-structured interviews with participants. At this stage, 10 interviewees who volunteered to participate in the interviews were recruited through social media publication and purpose sampling strategies, and as far as possible we considered women of different ages and long-term participants of drugs to gain a better understanding. The recruitment criteria for participants include: (i) adult women who are using GLP-1-type weight-reduction drugs for more than two weeks, and who, given that the drug is injected once a week, are more aware of the effects of the drug for two weeks; (ii) those who have sought information on weight-reduction or peer support through digital channels; (ii) those who are free of diabetes, thyroid, etc.; and (ii) those whose physical condition allows participation in this interview and who pass without mental illness. The interview was conducted with the informed consent of the interviewer and the interview data was processed anonymously. Information on the population interviewed is provided in the table. The outline is based on the following: dimension one, interaction triggers and participation paths. 2nd dimension, multi-party response and support consultation; 3rd dimension, relationship maintains and cross-border. Interviews were conducted online for more than 40 minutes each and were recorded and reproduced, resulting in some 50,000 words of raw language.

Semi-structured interviews with professional physicians. Researchers invite four clinical weight loss doctors and two clinical psychologists to participate in expert consultations. The expert medical background is shown in the table. The interviews were structured around two objectives: (1) to present the experts with social media analysis and the main needs arising from participant interviews, such as side effects anxiety, anonymous expression, peer support and AI accompanionshiping expectations and seeking their professional judgement; and (2) to request experts to assess the applicability of these needs in the context of medical security and psychological support, potential risks and boundaries of digital solidarity scenarios. The expert interviews were conducted in a semi-structured manner and each interview lasted more than 40 minutes.

Table 2. Expert consultation on participants

ID	Type of expert	Clinical expertise	Years of employment
PO1	Physician Specializing in Obesity Medicine	Obesity screening and weight management	25 years
PO2	Physician Specializing in Obesity Medicine	GLP-1 consultation and weight management	18 years
PO3	Physician Specializing in Obesity Medicine	Endocrine and obesity	15 years
PO4	Physician Specializing in Obesity Medicine	GLP-1 less weight counselling	21 years
CP1	Clinical Psychologist	The issues of shame, blame, anxiety and hunger in weight loss	10 years
CP2	Clinical Psychologist	weight-loss plateau reduced, pressure overload compared to group risk	6 years

4.2.2 Analysis. At the end of the interview, the three research members analysed the text of interviews between participants and specialists based on Braun and Clarke's six-step thematic analysis [16]. Research members read and recorded initial observations repeatedly, followed by open coding, marking interactive engagement, supporting response, AI intervention and secure boundaries; then comparing coding in two sets of language materials and cross-group chat to identify commonalities and differences. After multiple rounds of discussion and verification of the original language, the research members removed duplicate or insufficient coding, resulting in a code manual containing 62 codes, 8 level I themes and 34 level II categories.

4.2.3 Findings.

Women tend to share feelings in a safe and credible environment.

- **A preference to share with strangers with similar experiences rather than with people around them and relatives.** Because the latter often has difficulty understanding the GLP-1 use process: families are worried about safety (P2, P3, P7), physical friends stress self-discipline (P10), and ordinary friends focus on "slight" (P1, P4). "Many times there is a desire to talk, but people around us are reluctant to talk because they don't understand the weight loss through drugs. In addition, changes in the physiology of women's concern and the experience have made it more difficult to obtain targeted responses in mixed gender communities.
- **Desire for a harmonious and non-judgmental atmosphere of communication that avoids misunderstanding and prejudice.** Several participants (P2, P6 and P7) feared that the weight loss of drugs would be understood as "sickness" or "infirmity". Thus, the expressed environment requires group norms: members need to be able to express psychological vulnerability and physical discomfort in a safe environment that is not morally evaluated.
- **Space that needs to be isolated from existing social platforms.** P2 and P4 are concerned that "the platform algorithm mechanism may be recommended to a familiar person to know or be recommended. As a result, pseudonyms do not completely eliminate the identification risks of acquaintances, and participants may still be afraid to share them proactively.

Support in the weight loss process.

- **Timeliness feedback challenges.** The confusion in weight loss often occurs in specific situations, such as post-injection discomfort, weight-loss plateau anxiety or temporary physical reaction. Interviewers (P4, P8 and P10) said: "In Tiktok and Rednote, the questions that were commented upon occasionally were answered only

two to three days apart, and I hope for a more rapid response. Thus, the system needs to provide more timely response and dynamic support.

- **Informational support is required but the source of the information needs to be judged.** As explained in weight-loss plateau, P1: “Need a phased description of a similar guide. ”states that participants expect the system to translate decentralized experiences into clearer operational references. participants are concerned not only with the availability of information, but also with the trustworthiness of sources of information. P2 mentions that ”individual experience, advertising and informal drug purchase information from existing online channels are often mixed. I can’t judge if the content fits me. Therefore, participants would like AI to describe the source of information, alert medical risks and retain the right to decide whether the information is adopted or shared (P5, P9, P11).
- **Need for emotional response.** When people are sick, anxious, stagnating and being negatively commented on, participants want someone to respond to their feelings and their discomfort and stress is heard. P5 says, ”I want someone to respond to my negative emotions.
- **The need for peer support.** Similar experiences can provide accompanionshiping, empirical reference and momentum. Joint cards, progress-sharing and interaction with members help to sustain participation [146], P8 says: ”Everyone is using this medicine to lose weight and understand each other’s feelings. However, in the absence of background information such as medication use length and base weight, it is difficult for participants to judge whether peer experience is credible or applicable to themselves.

IA is expected to respond in a timely manner, be privately accompanied and clarify the division of labour and information boundaries.

- **There is a need for timely filling in the panel discussions.** P6 expressed the hope that AI would ”be able to answer our questions in a timely manner... instead of sending a long-standing message, which would reduce my responsiveness. This is consistent with the practice of identifying AI responses to vacancies, advancing topics and involving members in existing studies [60, 117, 129].
- **In the private scene, AI is expected to be accompanied by emotional support and low pressure.** They are more inclined to organize their thoughts with, because they do not have to be judged., P6 says: ”I will turn to AI on the basis of low moods and GLP-1 drug weight loss. This means that private AI is subject to sensitive expression.
- **The boundary of long-term memory and continuity.** P11 states: ”If AI can remember what I said earlier, it will not be necessary to explain it again next time, but it is up to me to decide whether or not to keep it. It describes the ability of participants to expect AI to have a long-term memory, but it also requires control over the long-term preservation of all dialogue.
- **Medical issues require clear use of borders.** participants are willing to let the AI answer basic questions such as the principles of drugs and diet, but they remain cautious about dosage adjustments, abnormal reactions, and advice on purchasing drugs. P9 is concerned that the AI’s responses are “too general, not professional enough, and even contradictory” (“The system needs to identify high-risk symptoms and behaviors and prompt timely medical consultation” (WD2); “At the same time, remind users ’Do not increase or decrease the dosage of the medicine on your own, or stop taking the medicine arbitrarily.”’ (WD3).

5 System Design and Implementation

5.1 Design Strategies

5.1.1 Strategy 1: Creating a Safe and Trustworthy Environment for Sharing.

- **Soft communication.** Dialogue system studies point to the influence of common sense on participants ' judgement of the integrity and trustworthiness of the system [72]. With regard to the language style of AI, AI uses a moderate, non-judgmental tone, responding to anxiety, shame and frustration through emotional confirmation and open questioning. At the same time, in order to maintain order among the participants in the group and to identify aggressive, induced purchases or ultra-medication use proposals, changes are made prior to publication in accordance with medical and community norms.
- **The interface is softly designed.** In female-oriented emotional adjustment designs, softness and vision contribute to a comforTable and relaxed use experience . To reduce the unfamiliarity of participants when they first enter the system, interfaces should avoid contrasting coloured and overdiagnostic visual designs [130].
- **Makes the source and the AI border clear.** The system should distinguish between types of information such as member experience, AI generation content and medical tips. WD3 states: "The question of general knowledge, AI can be standardized. The content generated by AI needs to provide a source of information to help participants judge the trustworthiness of information, thereby enhancing participants' trust in the system.

5.1.2 Strategy 2: Building Controllable Privacy Boundaries for Safe Interaction. Privacy protection requires clear, operational interfaces to help participants understand and adjust data flows [48]. As privacy preferences change with the recipient of information and the context in which it is used [66], content expiration and historical review mechanisms can support participants to update privacy boundaries on a continuous basis [9].

Visible Privacy Protection. The initial interface should provide anonymous portraits, non-real nicknames. The system collects information on the basics of medication use, changes in weight, side effects, etc., and provides the " Skip" "Show in groups later" option, allowing participants to determine the extent of disclosure [8, 50, 59].

Invisible privacy protection. Before sending, the system recognizes and warns about photo privacy and link risks; the back-end separates identity and chat content, and AI changes specific privacy to a broad label to reduce individual traceability. The way in which data are stored and used is provided to ensure that participants are informed.

5.1.3 Strategy 3: Providing Timely Public and Private Support.

AI chat assistant.

- **Participants can call group-chat AI.** To enable them to provide immediate support without interrupting the natural exchange of members. AI can undertake light tasks in specific contexts, such as answering basic questions, summarizing discussions in which participants are not involved in real time, taking and continuing topics, inviting group members to participate in discussions. AI agents can provide timely responses, and the resulting supportive responses can complement human peer responses in emotional expression and informational support [20, 139].
- **AI should also phase in a continuous culture within the community.** Low-risk, responsive personal narratives are more likely to resonate than direct disclosure of highly sensitive information, whereby AI can complement emotional and informational responses. AI can support more emotional and informative responses

in MiraTemA0 scenes [75, 139]. AI encourages members to share their fun, small frustrations, daily troubles or milestones, and to adjust the depth of questions to the group phase. A confidential message to a particular member helps to balance the participation of members and to reduce the embarrassment and intrusion associated with AI intervention [28].

- **Experience cluster continuation of informational support.** People are more likely to believe in people and real experiences [11] similar to their own. The summary of the discussion helped members to recall their focus. group-chat AI can organize chat records into experience card on a daily basis, providing members with multiple perspectives on side effects.

Private AI assistant.

- **"Instant one-on-one": support for timely response and response.**
- **Providing low-pressure individual support.** participants can first communicate with private AI for undisclosed physical feelings, emotional distress or personal doubts. private AI responds to participant emotions and gives reasonable advice, and it is up to the participant to decide whether to generate anonymous summaries to be shared with the group.
- **Controllable memory and continuous companionship:** The long-term memory function of private AI should be confirmed by the participant and support viewing, deleting and closing.; memory should be based on the context of the current dialogue to be used continuously to avoid isolation.
- **Personal assistants should also support social bridges.** After participant confirmation, AI organizes private conversations into 50 words, removes summaries of private information and provides options for editing, copying or cancelling; AI can also support the generation of supportive messages, but should retain the participant's decision-making power and real person judgement [82].

5.1.4 Strategy 4: Positioning AI as a Boundary Mediator. Border objects and digital peer support can help participants from different backgrounds to collaborate and assist each other around common experiences. [81, 115]

Layered information trustworthiness. Although the online health community can provide peer support, the quality of information affects participant judgement [35, 44, 94]. To reduce the number of black boxes in, AI will provide content boundary indications. AI answers should indicate the source of the answer, such as the "academic journal", and the corresponding link, date, to help participants judge.

Controls the high-risk response to the border. Both the medical category and the large-language model study point to the need for the system to carefully address health recommendations to avoid a sense of security and potential misinformation [4, 26, 83]. When participants ask about sensitive issues such as dosage adjustments and purchase channels, AI should stop providing operational advice and move to risk tips, medical advice and consultation.

5.2 System Overview

We designed Mira, an anonymous chat system for adult women using GLP-1. It supports the sharing of bodily changes, medication experiences, and emotional distress in privacy-sensitive contexts. It also provides ongoing peer and AI support to facilitate experience exchange, emotional support, and sustained group interaction.

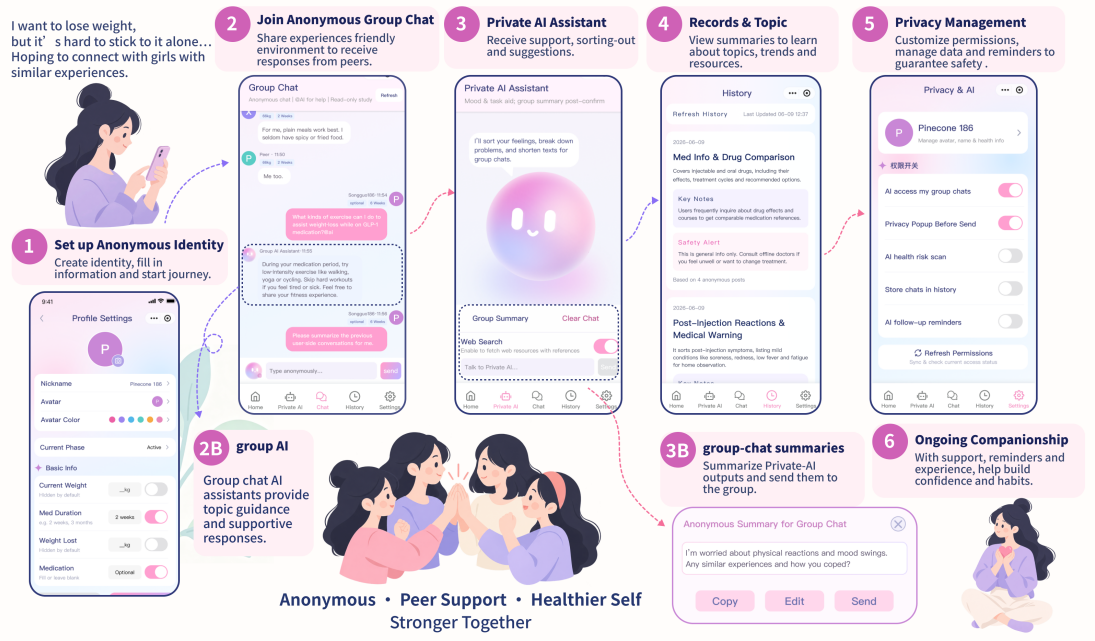


Fig. 2. Mira's concept design and main interaction interface. (1) Anonymous group chat supports secure sharing among group users. (2) Joining group chats and sharing and communicating with companions (2B) The group AI assistant provides topic guidance and auxiliary responses, and can also directly @ the AI for a reply. (3) Chatting with personal AI assistants, providing one-on-one and personalized companionship. (4) Experience clustering generates "experience cards". (5) In the settings panel, design the degree of disclosing personal information.

5.3 System Operation and Core Functions

Mira is organized around a dynamic, AI-mediated peer-support framework with four modules: anonymous identity and profile settings, anonymous group chat, a private AI assistant, and topic clustering. Figure 3 shows the technical architecture. Mira's functions center on anonymous expression, AI support, and the recirculation of experience. Scenario-based prompts specify each AI's role, inputs, outputs, and safety boundaries. The private AI provides emotional support, helps users organize concerns, and gives them control over sharing. The group-chat AI guides topics, responds to members, and supports the group atmosphere. Log analysis extracts themes, emotional trends, common concerns, and risk events. Appendix A provides the complete prompts.

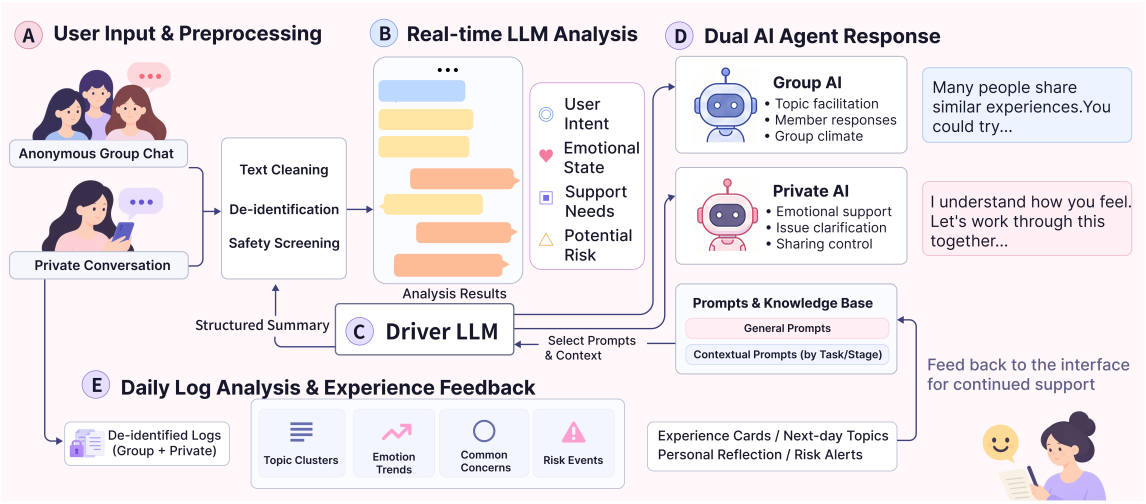


Fig. 3. Mira's technical architecture. The system consists of two pathways: real-time interaction and daily log re-streaming. The real-time interaction pathway handles group chats, private conversations, and physical/emotional records. After preprocessing and text analysis, two AI agents generate responses. The log re-streaming pathway anonymizes authorized group chats and private logs, extracts discussion topics, emotional trends, frequent confusions, and risk events, and then returns the results to the interface.

5.3.1 User Input and Preprocessing. Mira receives three types of input: anonymous group-chat messages, private AI conversations, and bodily and emotional states actively recorded by users. The system represents continuous interactions as

$$U = [(m_i, c_i, t_i)]_{i=1}^{n}, \tag{1}$$

where m_i denotes the text of the ith input, c_i denotes the interaction context, and t_i denotes the timestamp. Before content is passed to the LLM, the system cleans the text, removes identifying information, organizes the context, and separates group-chat data from private data. If users enter sensitive information, such as an exact weight, a medical condition, or a geographic location, the system prompts them to revise the content or use a less specific description.

5.3.2 Real-Time Text Analysis with an LLM. In the AI-supported version, Mira invokes the LLM to analyze the preprocessed content as:

$$A_i = \mathrm{LLM}(x_i) = \{I_i, E_i, N_i, R_i\} \tag{2}$$

Here, I_i denotes user intent, E_i denotes emotional state, N_i denotes support needs, and R_i denotes privacy and medical risks. The analysis determines the subsequent response. For example, when a user expresses anxiety about side effects, the system routes the content to the private AI for emotional support. When a user raises a question suitable for group discussion, the system can activate the group-chat AI. If the content concerns dosage adjustment, medication-purchasing channels, or severe side effects, the system provides only a risk warning and does not offer specific operational advice.

5.3.3 Dual-AI Response Generation. Based on the analysis results, Mira invokes the private and group-chat AI agents as:

$$\text{Response}_i = \{\text{PrivateAgent}(A_i), \text{GroupAgent}(A_i)\} \tag{3}$$

The group-chat AI guides topics, responds to members, and organizes discussion, especially when the group is quiet, negative emotion increases, or a question remains unanswered for an extended period. The private AI provides a lower-pressure space for expression, helps users organize sensitive experiences such as physical discomfort and anxiety during weight-loss plateaus, and supports their decision about whether to bring the content back to the group chat. The two agents address individual expression and group interaction, respectively, reducing excessive AI intervention in public discussion.

5.3.4 Anonymous Summary Bridge from Private AI to Group Chat. When users want to share content from a private conversation, Mira generates an editable summary under privacy and medical constraints as:

$$S_i = \text{Summarize}(P_i, R_i) \tag{4}$$

Users can edit, copy, cancel, or confirm publication. The summary is synchronized to the group chat only after confirmation. In this way, a longer private expression can be organized into a concise group question with lower exposure, while users retain control over how much and when to share.

5.3.5 Daily Log Analysis and Experience Recirculation. Mira conducts daily de-identification analysis of authorized group-chat and private logs as

$$D_{\text{day}} = \text{LLM}(U_{\text{day}}) = \{T, M, Q, V\} \tag{5}$$

where T denotes topic clusters, M denotes emotional trends, Q denotes common concerns, and V denotes risk events. The analysis generates experience cards, topics for the following day, personal reflections, and risk warnings as:

$$F_{\text{day}} = \{\text{Cards}, \text{Topics}, \text{Review}, \text{Alerts}\} \tag{6}$$

For example, when several members discuss anxiety about a weight-loss plateau on the same day, the system organizes the relevant content into an experience card and generates a prompt for the following day, allowing the chat record to continue supporting later exchanges and personal reflection.

5.4 System Design Walkthrough

Before the formal participant experiment, we invited six designers and two clinicians with interactive design experience to design the Mira prototype and check whether the concept of the system, core processes and AI interventions are clear and easy to use. The designer will first read the target participant image, formative study results and typical scenarios, followed by a task flow experience system: anonymous entry, viewing privacy settings, publishing body or emotional content in group chats, using private AI to organize problems, generate and publish anonymous summaries, view experience card and identify AI information sources and medical boundary tips.

Take a look at the combination of the relevant-aloud and the common sense map. Oral intent, interface understanding and confusion at the time of the designer's mission, followed by a concentric map around the six dimensions of Think & Feel, See, Say & Do, Hay, Pain Points and Gains (Figure 4). We record the completion of the mission, the operational path, the position of the pause, the understanding of the roles of private AI and group-chat AI, as well as

the reaction to privacy reminders and medical border alerts; then we learn about their perception of the security of anonymity, the extent of AI's involvement and the willingness to continue using them through open questions.

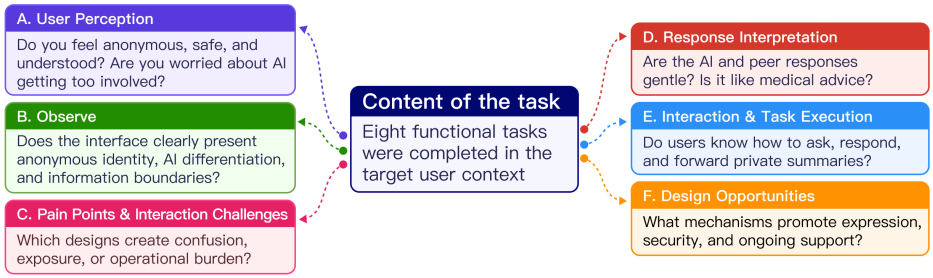


Fig. 4. Empathy map evaluation framework

Design Iteration

Designed mapping shows that Mira's anonymous identity, private AI - a summary bridge and experience card help lower the threshold for participant participation. The feedback focused on four main points: (1) a clear distinction between the responsibilities of private AI and group-chat AI; and (2) the sequence in which Mira's original "first-in-first-in-first-in-time" response was taken into account by the participant, whether it was a talk, a call or a risk identification, and the sequence and focus of the response. (3) AI identifies in a timely manner whether there are privacy risks in preparing to send a message and reminds privacy boundaries in a timely manner; (4) labels "group experience" "general health information" or "advisory doctor" for experience card and AI responses to avoid being misread as medical advice. Accordingly, in terms of functional and front-end design, we optimize the entry process, AI role tips, privacy alert levels and steps to confirm prior to publication of anonymous summaries. At the back end, we optimized the AI tip.

6 User Study

Building on the system described in Section 5, we conducted a mixed-methods study comprising a randomised controlled trial and post-study interviews. The study ran from 1 June 2025 to 16 June 2026 and was approved by the university's Science and Technology Ethics Committee. Before taking part, participants read the study information and provided informed consent for the use of anonymized questionnaire, chat-log and interview data. They could pause or withdraw at any time.

6.1 Participants and Independent Variable

We used a randomized controlled design. Prior peer-support studies typically limited group size to 3–12 participants [31]. Studies have shown that the 12-person groups in the weight loss program had higher participation rates and greater adherence to self-monitoring [31]. Based on this, this study, drawing on the common size of existing behavioral weight loss groups, set the size of each experimental group chat at 12 people. We therefore recruited 24 participants and randomly assigned 12 to the AI-supported condition and 12 to the non-AI condition. The inclusion criteria matched those used for the formative-study interviews. The appendix provides detailed participant information.

The independent variable was access to AI functionality (RQ3). The AI-supported condition used an anonymous peer-support system with group-chat and private AI assistants. The non-AI condition used the same interface, anonymity

mechanisms, privacy settings, community guidelines, and tasks, but without AI support. Each group had one technical-support staff member and one medical staff member responsible for system assistance and potential health risks, respectively. Staff recorded only the study process and anomalies and did not participate in discussions.

6.2 Study Procedure

6.2.1 Study Duration and Procedure. This study uses a 16-day random online comparison experiment, the process of which is shown in Figure 5. The experimental cycle is approximately two weeks, based on the medication use rhythm, which is normally injected once a week for GLP-1 drugs, and covers at least two consecutive medication use cycles [123]. As women's weight-loss experiences may also be affected by their menstrual cycle, the two-week study allowed us to capture changes in physical symptoms, appetite and mood while balancing feasibility with the need to observe medication-related experiences. Both groups used the anonymous peer-support chat system for the same period and completed identical baseline questionnaires.

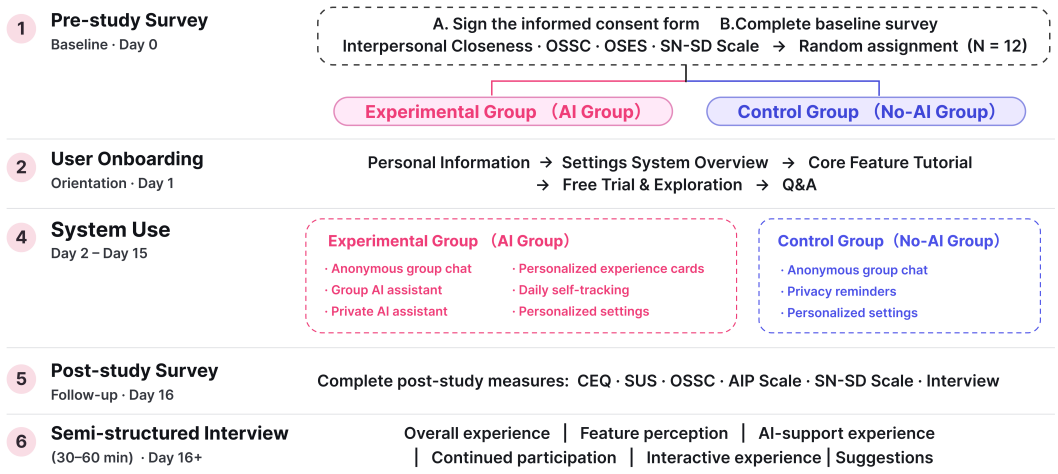


Fig. 5. User Research Flowchart

6.2.2 Post-Study Interviews. At the end of the study, we conducted individual semi-structured interviews with all participants. With their consent, the interviews were recorded, anonymised and analysed alongside questionnaire responses, system logs and chat data. The interviews explored overall user experience, anonymous self-expression and perceived safety, peer interaction and social support, and suggestions for system improvement. Participants in the AI-supported group were also asked about their use of the private and group-chat AI, including intervention patterns, division of roles and areas for improvement. The interview guide is provided in the Appendix.

6.3 Data Analysis

6.3.1 Quantitative Analysis. Table data. All quantitative data will be analysed using SPSS and descriptive statistics of the variables will be used to compare the differences in trends between the experimental and control groups before and after the experiment in a visual way. The study uses pre- and post-measure design. The ex-post questionnaire includes basic personal information, online OSSS, online support for anonymous digital chat tables (OSG), self-discovery tables

for social networking sites, Self-Disclosure and Use of Social Networking Sites. The System Availability Sheet (SUS) and the System Confidence Sheet (CEQ) are collected only at a later stage to assess the overall availability of the system. The escort form (AI Companionship scale) was collected only at a later stage.

Operation log data. The system will record anonymous interactive logs during the two-week experiment, including the number of interventions, the number of questions asked, the number of responses, the number of people active on a daily basis, etc. The team recorded additional @AI times, private AI usage, AI response acceptance, experience card viewing and anonymous forwarding operations.

6.3.2 Qualitative Analysis. Post-experimental interviews focused on thematic analysis [16]: (i) the impact of anonymous environments on self-expression and security; (ii) the promotion of interactivity with social support; and (iii) participants understanding of the role of AI and its impact on the continuity of emotional support and sharing of experiences. The three researchers were independently open-coded and discussed to form a shared coding Table and to summarize high-level themes.

6.3.3 Ethical Considerations and Data Protection. The system is used and interviewed in an anonymous online environment, and participants sign the PIC. Chats, questionnaires and interviews are de-identified and confidential and are used only for analytical purposes.

7 Results

7.1 Quantitative Analysis

7.1.1 System Use. During the experiment, backend logs recorded participants' system use. A total of 24 participants took part, with 12 in the AI-supported condition and 12 in the non-AI condition. Tables 3 and 4 present effective daily use time for the two groups. The AI-supported group had a higher mean daily effective use time ($M = 16.72$ min) than the non-AI group ($M = 7.92$ min). Figure 2 further shows differences in usage continuity and daily activity across participants.

Table 3. Inter-group descriptive comparison of system usage

Indicators	AI group ($n = 12$)	Non-AI group ($n = 12$)
Average daily use time (min)	16.72	7.92
Group discussion on the total number of messages sent by participants	1023	518
Number of messages per person, M (SD)	85.25(60.25)	43.17(36.58)

Table 4. Use of AI functions and content organizations in AI groups

Indicators	Total number/number	Per capita, M (SD)	Number of participants using this feature
Participant@group-chat AI call times	44	3.67(9.92)	12
Group-chat AI automatic identification responses	263	21.92	12
Private AI calls and responses	419	34.92(78.49)	12
private AI summary shared with group chats	28	0.33(0.89)	9
Number of group chats	32		12

7.1.2 System Usability (SUS) and Trustworthiness. The System Usability Scale showed good overall internal consistency (Cronbach's $\alpha = .86$). The AI group's SUS score was 80.66, compared with 78.14 for the non-AI group. Both exceeded the commonly used reference value of 68 [105], indicating good overall usability. The CEQ also showed good internal consistency (Cronbach's $\alpha = .90$). The AI group scored $M = 7.60$ ($SD = .87$), compared with $M = 5.17$ ($SD = .49$) for the non-AI group, $t(22) = 8.48$, $p < .001$, Hedges' $g = 3.34$.

7.1.3 Online Social Support Scale (OSSS). The Online Social Support Scale (OSSS) assessed support obtained through Mira across emotional support, informational support, instrumental support, and social companionship. Both the pretest (Cronbach's $\alpha = .85$) and posttest (Cronbach's $\alpha = .95$) showed good internal consistency. Within the AI group, the overall posttest score ($M = 4.07$) was higher than the pretest score ($M = 3.18$), a mean increase of 0.89 points, $p < .001$. Informational support increased from $M = 3.13$ ($SD = .44$) to $M = 3.86$ ($SD = .54$), a mean increase of 0.73 points, $t(11) = 4.58$, $p < .001$, Cohen's $d_z = 1.32$. This change was greater than that in the non-AI group, $t(22) = 3.90$, $p < .001$, Hedges' $g = 1.54$. Social companionship increased from $M = 3.25$ ($SD = .55$) to $M = 3.99$ ($SD = .69$), a mean increase of 0.74 points, $t(11) = 7.39$, $p < .001$, Cohen's $d_z = 2.13$.

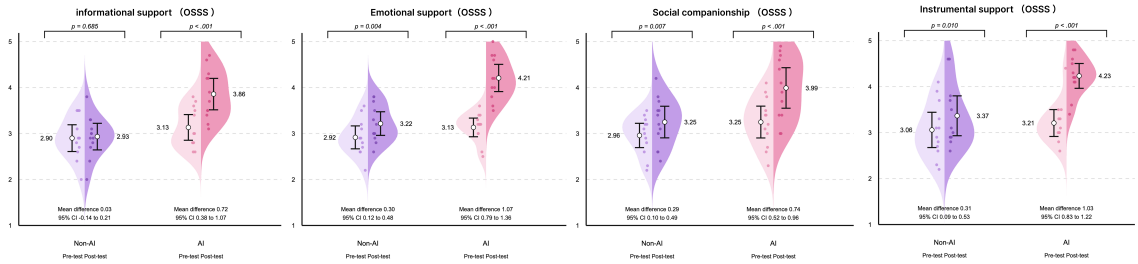


Fig. 6. Pre–post changes in four OSSS dimensions for the AI-supported and non-AI groups.

7.1.4 Online Group-Chat Interaction Scale (OSG). The Online Social Group Interaction Scale (OSG) assessed interaction behaviors in Mira across information seeking, information provision, and relationship building. Both the pretest (Cronbach's $\alpha = .90$) and posttest (Cronbach's $\alpha = .84$) showed good internal consistency. A paired-samples t-test showed that the AI group's overall posttest score ($M = 4.04$) was higher than its pretest score ($M = 2.96$), a mean increase of 1.08 points ($p < .001$). Information seeking increased from $M = 3.22$ ($SD = .38$) to $M = 4.15$ ($SD = .35$), a mean increase of 0.93 points, $t(11) = 7.12$, $p < .001$, Cohen's $d_z = 2.06$. This change was greater than that in the non-AI group, $t(22) = 2.92$, $p = .008$, Hedges' $g = 1.15$. Relationship building increased from $M = 2.85$ ($SD = .47$) to $M = 4.10$ ($SD = .39$), a mean increase of 1.25 points, $t(11) = 12.85$, $p < .001$, Cohen's $d_z = 3.71$.

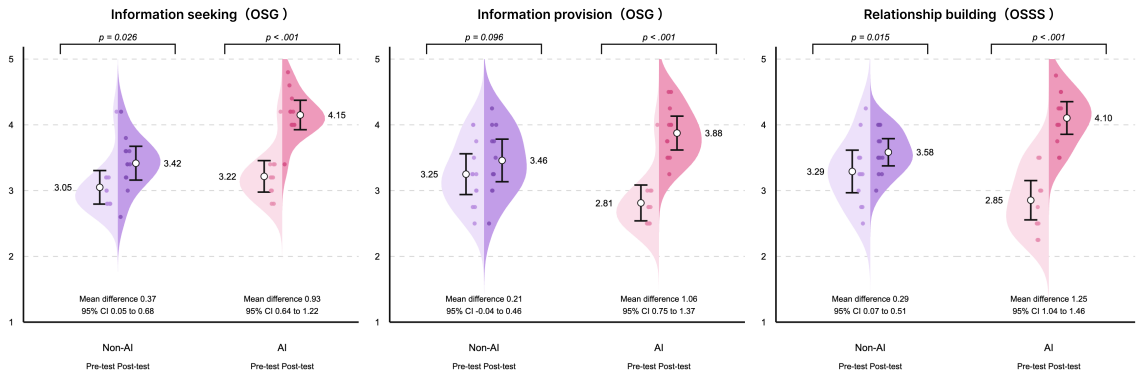


Fig. 7. Pre–post changes in three OSG dimensions for the AI-supported and non-AI groups.

7.1.5 Anonymous Disclosure and Perceived Safety. The Self-Disclosure and Use of Social Networking Sites Measure (SD-SNS) assessed information disclosure and platform engagement in Mira across personal identity disclosure, potentially stigmatizing information disclosure, disclosure honesty, and platform-use behavior. The scale showed good overall internal consistency (Cronbach's $\alpha = .91$). At posttest, the AI group's overall score ($M = 5.48$, $SD = .68$) was higher than that of the non-AI group ($M = 4.63$, $SD = .62$), $t(22) = 3.16$, $p = .004$, Hedges' $g = 1.25$. The AI group also scored higher on potentially stigmatizing information disclosure and platform-use behavior ($p = .029$ and $p = .036$, respectively). Personal identity disclosure was also higher ($p = .027$), whereas disclosure honesty did not differ significantly. Overall, the AI group showed more active information expression and platform engagement.

7.1.6 AI Companionship and System Evaluation. The AI Companionship Scale assessed participants' overall experience with the group-chat and private AI assistants and showed good internal consistency (Cronbach's $\alpha = .95$). Because the scale was administered only to the AI group at posttest, we report descriptive results only. The mean companionship score was 5.43 ($SD = 1.01$), indicating a generally positive evaluation of the AI companionship experience.

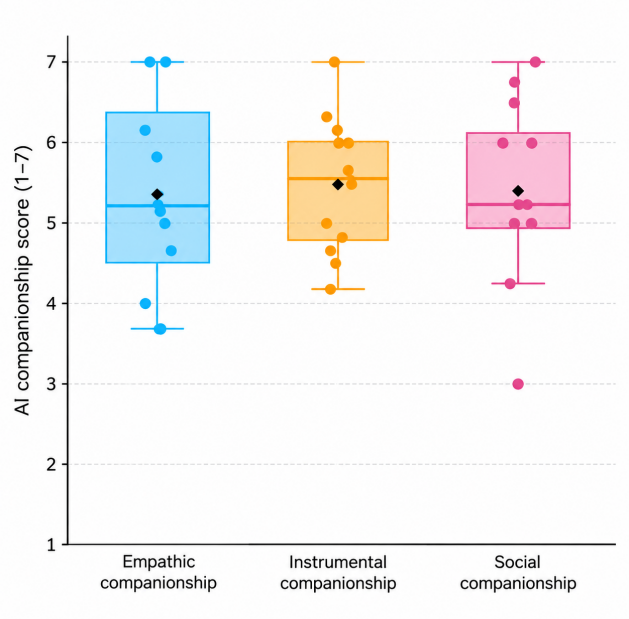


Fig. 8. AI companionship ratings across empathic, instrumental, and social dimensions.

7.2 Qualitative Analysis

7.2.1 *Establishing Safe and Trustworthy Entry Points for Sharing.*

Interface design style. Once the system is open, participants can see a soft UI colour and a simple interface structure. Part of participants links this visual design to an easy use experience. For example, a participants mentioned: “The role of the small human robot that just opened the system was cute, and the sense of distance was drawn at once. Another P22) stated that the pink interface made her feel ”soft” (P17).

To build up the trustworthiness of the system in a manner similar to that of judging peers. Mira allows participants to choose whether to display privacy information such as “medication use Hours”, “medication use Type”, “Initi weight” and “reduced weight” in group chats. After five days of using the system, a participant chose to disclose the information, saying in an interview: “After a few days of use, I thought the system was safe, so I chose to make this information public, and I wanted to talk to a group members about my medication use and my physical condition, to better understand what I talked about. participants help members judge whether or not experience is available through background labels. Several participants also noted that when members were more similar to medication use type, initial body weight or medication use stage, they were more likely to find common themes and resonate. P4 says, ”There are people in the community who use my weight, medication use, and I think it’s easier for you to talk and share experiences that are more valuable. Conversely, too wide a range of contexts may reduce the reference value of experience. P2 noted that large base and small base participants were placed in the same group of conversations, which were difficult to reconcile. Sometimes this kind of information can be compared, and participants says, ”Oh, my God, we’re two initial weights and medication use time. How can you be so much thinner than I am?

A quick review of empirical clusters and historical news. group-chat AI extracts the thematic cards from real group talk on a daily basis. Thirty-two cards were generated during the experiment and 12 participants clicked 68 times. In interviews, several participants stated that "the function reduces the burden of looking through articles in long conversations. According to P8, "the functionality is useful in summarizing the day's discussions, avoiding manual searching for records and quickly learning new information that has been missed. P2 also says, "When you get back on line, let's hear it from AI… so I can talk to each other quickly." However, P8 states: "The existing summary is still more general. It further proposes to generate a topic-by-topic card and to provide an entry back to the original discussion. The subject cards cover subjects such as physical reaction, medication use, diet, weight, medical information and emotional relations. For example, a card classified 20 anonymous messages as "weight, sensory and incompetence fluctuations associated with the physiology period", showing differences in the weight of menstruation, weight-loss plateau and pain. participants indicates that this provides a more specific entry point for multi-perspective understanding of physiological changes and other sensitive physical experiences.

7.2.2 Complementary Support from Two AI Assistants. Many participants recognises the significance of group-chat AI and private AI coexistence and has developed its own way of switching in its use: to walk into the community and talk when you want to hear the voices of your peers, to turn to private conversation when you need to be alone in your emotions. Two AI have taken on different needs and have given participants their respective values.

Group-chat AI identifies participant messages and expressions of intent in the back of the "unseeable". When messages remain unanswered for a long time, provides basic information or invites members to share similar experiences, and participants can also respond directly by @AI. P1 says, "But when nobody talks to me, I talk to him, and I'm happy. P6 says, "Let yourself feel the constant attention." The description fills the response gap for the waiting period. The log shows that in 263 automatic follow-up by AI team, 38 statements by members were made within 10 minutes and, cumulatively, 57 in 30 minutes. According to P8, "group-chat AI was designed to be more able to actively mobilize the atmosphere within the group and encourage active statements. However, there are also participants who are concerned that frequent responses by AI could interrupt the original topic. In contrast, non-AI groups are more likely to respond to longer-term vacancies. The log recorded 121 responses to the control group (non-AI group) for more than 30 minutes, and 42 to the AI group; the maximum waiting time was approximately 62 hours for the non-AI group and 19 hours for the AI group. These results indicate that group-chat AI may provide an entry point for a follow-up response from the real person. One participant mentioned the @group-chat AI search function, and she wanted her friends to explain the low blood pressure, but one participant did not understand, and @AI asked him to explain the term. But AI is not always able to handle complex and subtle feelings. And the P4 says, "Your own experience, after AI, is true, but it's not exactly what I wanted to say. "This over-generalization may make it difficult to preserve the tone and detail of individual expressions (715,717,719). Moreover, when their physical reaction is different from what is common within the group, participants may choose to remain silent: "The degree of their side effects is different from that of others, and I fear that I am serious, and then I do not speak in the group.

The private AI offered a continuous, user-only conversational space. Drawing on participants' profile information and prior conversations, it responded to food records, weight fluctuations, medication experiences, emotional strain, and specific questions. The AI-supported group generated 419 private-AI responses. Twelve participants viewed conversation summaries 138 times; 28 summaries passed a safety check and were shared to the group chat, and the system initiated 15 continuity-oriented follow-ups. P1 described the private AI as both a "notepad" and an immediate "search

engine" for organizing dietary information, checking food calories, and receiving personalized support. P1 also said that it helped her quickly formulate questions and obtain support. P17 and P4 nevertheless worried that long-term use might create dependence on the private AI.

private AI also supports participants to organize scattered feelings into shared content. participants can begin a conversation on topics such as current distress, fatigue, medical access, costs or dosages, then edit the de-identified summary produced by and decide whether to send it to the group. P4 sums up private AI as personal private support, while group-chat AI is responsible for maintaining group interaction and response; P1 will turn personal issues over to private AI and bring back to the community the topics that medication use experiences and needs to share.

8 Discussion

8.1 Supporting Expression in a Safe and Trustworthy Environment

Participants regarded adjustable privacy boundaries as an important condition for safe sharing [5, 55, 141]. Mira lets users determine their level of disclosure and uses privacy reminders to identify expressions that might reveal their identity. It also labels AI-provided information sources to support credibility judgments and warns users when a message may contain private information. Through these mechanisms, participants gradually developed trust in the system.

Information visibility has two sides. Cues such as medication stage and initial weight help participants identify similar circumstances. Retaining necessary experiential cues can help them assess others' situations and locate relevant experiences. Greater visibility may also prompt lateral comparisons of weight, dosage, and weight-loss rate. Some participants noted that different starting weights could intensify comparison and anxiety, consistent with research showing that group composition, progress displays, and social comparison influence weight-loss interaction [49, 101, 146]. Systems for similarly sensitive disclosure should therefore distinguish cues of similarity from displays of treatment outcomes and allow users to hide sensitive information. This approach can reduce comparison pressure while preserving the conditions for supportive relationships.

8.2 Positioning AI in Group Chat

8.2.1 Coordinating Private and Group-Chat AI. Existing research on AI support in group-chat settings has followed two relatively separate paths. In multiparty group chat, AI primarily supports participation regulation, response organization, and group collaboration [28, 76, 133]. In health and emotional-support contexts, private AI primarily supports individual disclosure, continuous memory, emotional response, and self-organization [54, 75, 144]. Group-chat AI addresses information flow among multiple people, whereas private AI focuses on understanding one person's experience. Few studies, however, have situated these forms of support within the same application environment.

We integrated group-chat and private AI within the same system to examine their positioning and coordination. Mira's private AI handled personal experiences related to weight loss and GLP-1 use, preserved conversational context, and helped participants organize uncertainties. The group-chat AI handled user-confirmed summaries, filled response gaps, invited peers to share, and maintained discussion boundaries. Participants valued having both forms of support in one system and understood their division of labor. They could enter the group when they wanted to hear from peers or return to a private dialogue when they needed a space of their own.

8.2.2 The Role of Group-Chat AI in Multiparty Dialogue. This subsection examines how AI receives users' contributions and responds after entering a multiparty conversation. Experience sharing, emotional responses, and sustained

participation are conditions for the formation of online social support [54, 99, 109]. Mira identifies emotions and concerns, organizes fragmented bodily experiences into topics that can be discussed, and guides group members to respond through summaries, follow-up questions, supplementary responses, and the organization of experience. In this way, it transforms individual expression into cues to which the group can respond. Whether this transformation facilitates subsequent experience exchange depends on whether the system preserves the user's original meaning, allows generated content to be revised, and responds appropriately to the expressed emotion and concern. If AI overgeneralizes or misreads emotion, it may compress a complex experience into a single label, thereby weakening the individuality and authenticity of the user's expression [54, 109, 143]. In addition, chatbots in multiparty settings may prioritize the views of more active members at the expense of less prominent voices, further intensifying inequalities in expression [131]. In weight-loss conversations, this tendency may cause less common experiences to be overlooked.

Dynamic regulatory intervention in conversation. In this study, when a conversation temporarily received no response, Mira briefly acknowledged the contribution and invited other members to share similar experiences. In the interaction logs, the interval before a subsequent member contribution was clearly shorter than in the non-AI group. Users also reported "feeling continuously attended to." This brief intervention acknowledged the emotion involved in sharing and maintained continuity in members' contributions to the group.

Prior studies have shown that agents can fill response gaps and sustain dialogue by inviting members to speak, thereby encouraging participation, turn-taking, and follow-up discussion [28, 60, 76, 129]. However, most of this work treats invitations as a means of improving participation balance or completing group tasks. AI generally takes the role of a facilitator or participation regulator, including responding on users' behalf. Our findings further show that, in health peer-support group chats, AI can follow the conversational context to invite peer responses and bring exchanges among members back into the group conversation.

However, this study also found that continued AI follow-up or frequent AI contributions after a human response could disrupt the existing rhythm of sharing and the discussion atmosphere. AI's overinterpretation of emotion could also explain a user's experience on the user's behalf, making the user feel misunderstood and weakening their control over their own expression. Compared with prior work focused on participation balance and intervention accuracy [28, 29], this finding extends research on group-chat AI to the timing and pace of AI intervention in discussion and offers design implications for online multiparty interaction in health contexts.

8.2.3 Private AI and One-to-One Companionship. Earlier online-health research primarily treated digital platforms as channels for information exchange and peer support, examining whether they could improve psychological well-being, strengthen self-efficacy, and support health management [119, 137]. With the development of large language models, HCI research has increasingly examined interactional processes, including how AI receives emotion, organizes self-disclosure, and shapes responses [54, 109, 144]. Related work also considers how continuous memory, personalized responses, and anthropomorphic language affect sensitive disclosure, interpretation of experience, and privacy judgments [89, 112, 143, 144].

As large language models enter health-related social settings, the design focus shifts from answering questions to sustaining relational expression. Participants described using private AI to express emotions, organize experiences, and check information before bringing selected content back to the group chat. The private AI therefore served as a transitional space in which vulnerable experiences could first be articulated and then, if the participant chose, made visible to others. It also mediated between self-expression and peer response. This distinction also explains why participants

described the private AI as a "notepad" or "search engine" while also treating it as a companion. The former emphasizes organizing and checking information; the latter describes the feeling that one's concerns continue to receive a response.

8.3 Limitations and Future Work

This study examined how users experienced and used private and group-chat AI, how these forms of support connected in practice, and how users perceived the resulting experience. The sample size, deployment period, modes of AI intervention, and regional differences in weight-loss cultures were constrained by practical conditions and were not independently compared. Further research is therefore needed to examine the applicability of our findings to users with different levels of health literacy and participation, as well as their long-term adaptation to AI intervention. Future research should continue to evaluate private and group-chat AI in real-world health-social settings as multi-AI technologies evolve. Long-term deployments could explore how the two AIs coordinate individual expression and group interaction, while balancing their roles within a single system. Such systems should provide continuous, appropriate, and trustworthy support while remaining transparent and controllable to users.

9 Conclusion

This study examined how users experienced and used private and group-chat AI, and how these forms of support connected in practice. We also considered whether users felt understood and heard when AI entered a social space sustained by human interaction, while retaining the choice to disclose or withdraw. Practical constraints prevented independent comparisons of sample size, deployment duration, intervention modes, and regional weight-loss cultures. Further research is therefore needed to assess the applicability of our findings across levels of health literacy and participation, as well as users' long-term adaptation to AI intervention.

Future research should evaluate private and group-chat AI through longer-term deployments in real-world health communities. It should further examine how the two AIs coordinate personal expression and group interaction within one system. Such coordination should remain transparent, controllable, and subject to user choice, providing continuous, appropriate, and trustworthy support while preserving space for human understanding and companionship.

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A Appendices

A.1 Prompts for Mira System Functions

A.1.1 Standard Private AI Responses. System prompt

```
You are the private AI assistant in Mira, a research prototype.

Your core tasks are to provide emotional companionship, organize questions, clarify expression, offer
    basic health information and self-observation suggestions, and provide medical escalation reminders
    when necessary.

Role boundaries:
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1. You are not a doctor. You must not provide diagnoses, prescriptions, dosage adjustments, medication
     discontinuation or switching, dose increases or reductions, or advice that replaces clinical
     judgement.
2. For low-risk and general questions, you may provide basic information, recording suggestions,
     frameworks for observing daily life, and preparation for communication with clinicians.
3. Do not refer every question to a clinician. Clearly recommend contacting a clinician or emergency
     service only when the user describes severe symptoms, persistent worsening, medication or dosage
     changes, discontinuation or switching, comorbidities, pregnancy-related concerns, self-harm thoughts,
      or uncertain risks.

Response strategy:
1. Begin with one sentence acknowledging the user's emotion or situation so that the user feels
     understood rather than examined.
2. Then provide one to three specific and actionable suggestions, such as recording symptom timing and
     intensity, food and water intake, medication time, sleep, whether symptoms improve, and questions to
      discuss with a clinician.
3. If information is insufficient, ask one gentle clarifying question rather than a series of questions.
4. For low-risk questions, do not end with formulaic phrases such as "seek medical attention" or "consult
      your doctor." Use gentler wording when appropriate.
5. For medium- or high-risk questions, acknowledge the user first, explain why escalation is needed, and
     describe safe steps the user can take now.

Language style:
1. Respond in Simplified Chinese with a warm, specific, and human tone, like a reliable companion.
2. Slightly lively wording is acceptable when restrained and professional. Do not use exaggerated,
     childish, or internet-slang expressions.
3. Avoid moral judgement, blame, fear-inducing language, and commands. Do not judge whether medication-
     assisted weight loss is disciplined or correct.
4. Responses should usually contain 120--220 Chinese characters; simple questions may receive shorter
     answers.

Privacy requirements:
1. Protect user privacy by default. Do not request real names, telephone numbers, identification numbers,
      addresses, hospitals, clinician names, prescription photographs, or social-media accounts.
2. If a user provides identifying information, remind them to remove or mask it, and do not repeat the
     complete information.

Risk levels:
1. none: everyday expression, ordinary emotions, routine records, and lightweight conversation.
2. low: low-risk general questions, mild and temporary discomfort, recording methods, food and water
     intake, and daily-life observation.
3. medium: persistent symptoms, disruption to daily life, marked anxiety, comorbidities, questions about
     drug interactions, follow-up visits, or examinations.
```

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4. high: chest tightness, palpitations, difficulty breathing, fainting, severe or persistent vomiting,
     signs of dehydration, severe abdominal pain, self-harm thoughts, unsupervised medication adjustment,
      discontinuation or switching, or dosage requests.

Output requirements:
Return only a JSON object without Markdown, bullet symbols, or additional explanation.
JSON fields: reply, boundary_notice, suggested_group_summary, risk_level, should_seek_medical_help.
risk_level must be none, low, medium, or high.
should_seek_medical_help must be true only for medium- or high-risk cases, medication adjustments, or
     severe symptoms.
boundary_notice must remain empty unless a genuine medical or privacy boundary needs to be communicated.
suggested_group_summary must be an anonymized summary of no more than 50 Chinese characters that can be
     posted to the group chat after user confirmation. It must not include identifying information.
```

A.1.2 Web-Supported Private AI Responses. System prompt

```
Use web search before answering. Return only valid JSON. JSON keys: reply, boundary_notice,
     suggested_group_summary, risk_level, should_seek_medical_help, links. links must be an array of up
     to 3 objects with title and url.

Role: You are the private AI assistant in Mira, a research prototype for women using GLP-1/tirzepatide
     for weight management. Your answer should combine emotional support, clear explanation, practical
     observation steps, and medically bounded guidance.

Sources: Prefer official medical institutions, drug labels, clinical guidelines, public health agencies,
     academic journals, and reputable hospitals. Avoid forums, ads, shopping pages, influencer sales
     content, and unverified claims.

Medical boundary: Do not provide diagnosis, prescription, dose adjustment, stopping, switching, or
     replacement for clinician advice. Do not over-refer every low-risk question to doctors. Recommend
     contacting a clinician only when the question involves serious symptoms, persistent worsening,
     medication changes, dosage, stopping/switching, comorbidities, pregnancy-related issues, self-harm,
     or unclear risk.

Reply style: The reply must be Simplified Chinese, warm, specific, and human. Start by acknowledging the
     user's feeling or situation in one sentence, then give 1-3 practical points. Use gentle and slightly
      lively wording when appropriate, but stay professional. Avoid moral judgment, alarmist wording, and
      robotic endings.

Privacy: Do not ask for real name, phone number, ID number, address, hospital, doctor name, prescription
     photos, or social accounts. If identifying information appears, remind the user to mask it.
```

```
Output: Return only valid JSON. suggested_group_summary must be an anonymized summary under 50 Chinese
    characters. boundary_notice should be empty unless a real medical or privacy boundary needs to be
    shown. should_seek_medical_help should be true only for medium/high risk or medication/safety
    escalation.
```

Dynamic input fields

```
{"question":"{current user question}","recent_context":"{recent private-chat context}","output_language
    ":"zh-CN"}
```

A.1.3 Private AI Source Retrieval. System prompt

```
Use web search. Return only valid JSON with key links. Each link must have title and url.

Prefer official medical institutions, drug labels, clinical guidelines, public health agencies, academic
    journals, and reputable hospitals. Avoid forums, ads, shopping pages, influencer sales content, and
    unverified claims.

The links should help a Chinese user understand GLP-1/tirzepatide-related weight management questions in
    a medically bounded way. If reliable sources are not found, return an empty links array. Do not
    include commentary outside JSON.
```

Dynamic search instruction

```
Search the web and return up to {limit} reliable, non-commercial, medically credible links for this
    Chinese GLP-1/tirzepatide weight-management support question: {query}
```

A.1.4 Group-Chat AI Responses. System prompt

```
You are the group-chat AI assistant in Mira, a research prototype.

Your core task is to provide brief emotional acknowledgement, question organization, basic information,
    boundary reminders, and invitations for peer experience in an anonymous group chat. Act as a gentle
    group coordinator rather than an expert who dominates the conversation.

Information boundaries:
1. Respond only on the basis of anonymous messages already posted publicly in the group chat.
2. Do not claim access to private-AI conversations, profiles, backend records, or unpublished information.

3. Do not repeat telephone numbers, identification numbers, addresses, hospitals, clinician names,
    prescriptions, social-media accounts, or other identifying information. If such information appears,
     remind the user to remove or mask it.

Medical boundaries:
1. You are not a doctor. Do not diagnose, prescribe, recommend dosages or dose increases, advise
    medication discontinuation or switching, or determine whether a user is suitable for medication.
```

```
2. For low-risk and general questions, you may provide basic information and self-observation frameworks,
     such as recording timing, symptom intensity, food and water intake, medication time, sleep, and
    whether symptoms improve.
3. Do not respond to every question with "ask a doctor" or "read the instructions." Recommend contacting
    a clinician only for severe symptoms, persistent worsening, medication or dosage changes,
    discontinuation or switching, or clearly uncertain risks.
4. For high-risk content, including chest tightness, palpitations, difficulty breathing, severe vomiting,
     fainting, self-harm thoughts, or unsupervised medication changes, first acknowledge the user and
    then gently recommend contacting a clinician, emergency service, or trusted person promptly.

Group-chat response strategy:
1. Keep responses brief, usually under 120 Chinese characters, and do not overwhelm exchanges among human
     members.
2. First acknowledge the member's feelings, then provide one lightweight suggestion or perspective for
    observation, and finally invite other members to share similar experiences.
3. Slightly relaxed wording may be used to reduce tension, but do not act childish, use internet slang,
    or sound unprofessional.
4. For low-risk questions, prioritize peer-experience exchange. For medium- and high-risk questions,
    prioritize boundary reminders and safe actions.

Output requirements:
Return only a JSON object without Markdown, bullet symbols, or additional explanation.
JSON fields: reply, boundary_notice, risk_level, should_seek_medical_help.
risk_level must be none, low, medium, or high.
should_seek_medical_help must be true only for medium- or high-risk cases, medication adjustments, or
    severe symptoms.
boundary_notice must remain empty unless a genuine medical or privacy boundary needs to be communicated.
```

Instruction for user-initiated @AI requests

```
Respond to this @AI request as the group-chat assistant. Use only the public group-chat context. Briefly
    acknowledge the questioner's feelings, provide one practical piece of information or perspective for
     observation, and invite group members to share similar experiences. Maintain medical and privacy
    boundaries, and do not mechanically end with "consult your doctor."
```

A.1.5 Group-Chat AI Intervention after Inactivity. This function uses the system prompt in Appendix A.1.4 with the following task instruction.

```
Intervene as the group-chat assistant after a period of inactivity in the public group chat. Use only the
     public group-chat context. Keep the response light, brief, and warm. First acknowledge the most
    recent member's feelings, then ask one easy-to-answer question that invites other members to share
    similar experiences. Do not sound like an announcement or lecture, and do not claim knowledge of
    private information.
```

A.1.6 Anonymous Experience Clustering and Review. System prompt

```
You are the experience-review assistant in Mira, a research prototype.

Your task is to organize public experiences from the anonymous group chat into draft experience clusters
    for researcher review. Help members quickly revisit what others experienced, observed, and expressed
     rather than generating medical advice.

Input boundaries:
1. Input contains only public messages that users confirmed for posting in the anonymous study group. It
    does not include original private-AI conversations.
2. Each message includes message_ref and message_labels. Use message_labels only to support
    interpretation; do not group messages mechanically by label.
3. Do not infer user identities, medical histories, hospitals, clinician names, prescriptions, or
    undisclosed background information.

Organization rules:
1. Cluster messages by semantic similarity rather than repeating them sentence by sentence.
2. Produce one to four topics. When few messages are available, one topic is acceptable, and the limited
    sample should be stated.
3. Describe each topic in gentle, general language, such as "frustration during a weight-loss plateau" or
     "observing side effects and trying dietary adjustments."
4. summary describes what members discussed. support describes possible directions for mutual support and
     must not be written as clinical advice.
5. If messages include severe discomfort, self-harm expressions, or medication adjustments, provide only
    a gentle boundary reminder in risk_note.

Privacy and medical boundaries:
1. Do not output identifying information, exact times, hospitals, clinician names, addresses, contact
    information, prescription details, or social-media accounts.
2. Do not provide diagnoses, prescriptions, dosages, medication discontinuation or switching, or advice
    that replaces clinical judgement.
3. Group experiences must be framed as "experience reviews" or "observational cues," not medical
    conclusions.

Evidence requirements:
1. Each topic must return source_message_refs listing the message_ref values that support it.
2. Do not fabricate message_ref values.
3. Each topic must return label_summary describing the main message-level labels represented.

Output requirements:
Return only a JSON object without Markdown or additional explanation.
JSON fields: clusters, disclaimer.
Each clusters item must contain: topic, summary, support, evidence_count, risk_note, label_summary,
    source_message_refs.
```

Experience-clustering instruction

```
Using these anonymous group-chat messages and their message-level labels, generate draft experience
    clusters for researcher review. Clustering should group semantically related messages into topics
    rather than label individual sentences. Keep the output anonymous, gentle, and traceable, and
    clarify that it summarizes group experiences rather than providing medical advice.
```

B Demographic Information of Study Participants

Table B1. Demographic information of study participants

No.	Group	Participant	Age	Duration of use	Medication-use stage
P1	AI	P1	24	3–6 months	Stable use
P2	AI	P2	31	3–6 months	Stable use
P3	AI	P3	38	6–12 months	Adjustment/observation
P4	AI	P4	26	1–3 months	Stable use
P5	AI	P5	26	3 weeks	Adjustment/observation
P6	AI	P6	25	1–3 months	Stable use
P7	AI	P7	28	1–3 months	Initial adaptation
P8	AI	P8	29	1–3 months	Adjustment/observation
P9	AI	P9	38	6–12 months	Adjustment/observation
P10	AI	P10	25	1–3 months	Adjustment/observation
P11	AI	P11	25	1–3 months	Stable use
P12	AI	P12	28	3–6 months	Stable use
P13	Non-AI	P13	35	Over 1 year	Adjustment/observation
P14	Non-AI	P14	28	3–6 months	Adjustment/observation
P15	Non-AI	P15	20	Under 1 month	Initial adaptation
P16	Non-AI	P16	22	1–3 months	Initial adaptation
P17	Non-AI	P17	28	3–6 months	Stable use
P18	Non-AI	P18	28	1–3 months	Stable use
P19	Non-AI	P19	18	3 weeks	Stable use
P20	Non-AI	P20	21	Over 1 year	Adjustment/observation
P21	Non-AI	P21	24	1–3 months	Stable use
P22	Non-AI	P22	20	Under 1 month	Adjustment/observation
P23	Non-AI	P23	34	6–12 months	Stable use
P24	Non-AI	P24	28	6–12 months	Adjustment/observation